AI TRADING AGENTS HACKATHON 2026

Proof-of-Trust Trading Agent

AI that can't break its own rules

ERC-8004 on-chain identity • 7 enforced risk gates • EIP-712 signed intents • Cryptographic audit trail

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THE PROBLEM

AI trading agents are black boxes

No Identity

Who is this agent? Who controls it? No way to verify.

No Audit Trail

When something goes wrong, you can't prove what happened or why.

No Enforced Limits

Risk controls exist in code the AI can override. Trust the operator, not the math.

Users trust opaque code with their capital and hope for the best.

OUR SOLUTION

The Trust Triangle

Every trade must satisfy all three — simultaneously.

1. Identity

ERC-8004 on-chain identity. Agent registered as ERC-721 NFT. Discoverable via Agent Card.

2. Constraints

Risk Router contract enforces 7 gates. The AI proposes — the system decides. No bypass possible.

3. Auditability

Every intent signed with EIP-712. Validation artifacts hashed on-chain. Cryptographic proof, not logging.

ARCHITECTURE

Hexagonal Design — Ports & Adapters

```
ERC-8004 Identity Registry (on-chain ERC-721)
    ↓
AI Strategy Engine (Momentum + Claude/ADK Hybrid)
    ↓
Intent Builder + Signer (EIP-712 typed data, secp256k1 ECDSA)
    ↓
Risk Gates (7 gates: drawdown, position, confidence, fee, regime, circuit breaker, ATR stop)
    ↓
Execution (Kraken CLI — paper or live)
    ↓
Validation & Reputation (artifact hash → on-chain reputation registry)
    ↓
Dashboard (SvelteKit) + HTTP API (/logs, /metrics, /agent-card.json)
```

TECH STACK

Production-Grade Infrastructure

Agent Core	Rust (async tokio, hexagonal arch)
AI Engine	Anthropic ADK-Rust (Claude Sonnet)
Execution	Kraken CLI (paper + live)
Signing	EIP-712 ECDSA (secp256k1)
Contracts	3 Solidity (Identity, Reputation, Risk Router)
Dashboard	SvelteKit + Chart.js
Chain	Base Sepolia (EVM L2)

70+
PASSING TESTS (RUST + SOLIDITY)

45+
RUST SOURCE FILES

3
SMART CONTRACTS

CORE INNOVATION

7 Risk Gates — AI Cannot Bypass

GATE	DEFAULT	ENFORCEMENT
Max Drawdown Cap	5%	Hard stop — agent pauses if equity drops from peak
Position Limit	1 open	No pyramiding — one position at a time
Confidence Floor	60%	Reject low-conviction AI signals
Fee Filter	0.7% edge	Don't trade unless move covers round-trip fees
Regime Filter	Auto	Hold during market transitions
Circuit Breaker	3 losses / $5	Auto-pause after consecutive losses or daily limit
ATR Trailing Stop	1.5x ATR	Mechanical exit — no AI discretion

20%+ of trades blocked — proof that gates are active, not decorative.

LIVE DASHBOARD

Real-Time Monitoring & Audit

- Price chart with Bollinger Bands (20,2)
- Buy/Sell trade markers on chart
- PnL tracking (realized + unrealized)
- Drawdown monitoring with circuit breaker status
- Full trade log with risk gate decisions
- Metrics bar: ticks, executed, blocked, holds, errors

HTTP API Endpoints

`GET /metrics`

Real-time counters

`GET /logs`

Full trade history with artifacts

`GET /.well-known/agent-card.json`

ERC-8004 discovery endpoint

ON-CHAIN TRUST LAYER

ERC-8004 — Full Implementation

On-Chain (Verifiable)

- Agent identity (ERC-721 NFT)
- Risk policy summary
- Performance metrics (Sharpe, drawdown)
- Validation artifact hashes

Off-Chain (Performant)

- Strategy logic (Rust core)
- Real-time indicator computation
- Trade execution (Kraken CLI)
- Dashboard + HTTP API

The agent works without the chain. The chain makes it verifiable and discoverable.

WHY THIS MATTERS

Trust > Returns

The winning AI trading agents won't be the ones with the best returns — they'll be the ones you can verify and trust.

- **Regulatory-aligned** — "constrained agent in a hard risk envelope" matches what regulators recommend
- **Auditable** — every decision has cryptographic proof
- **Trustless** — trust the math, not the operator
- **Discoverable** — Agent Card enables a marketplace of verified agents
- **Extensible** — hexagonal arch = swap venues, chains, strategies
- **Production-ready** — not a demo, a deployable system

THANK YOU

Proof-of-Trust Trading Agent

AI that can't break its own rules

GitHub
github.com/zixelfreelance/erc8004-trading-agent

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